Supplementary Information

Structure-predicted solvent responses enable simulation-free hydration free-energy prediction

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Supplementary Methods

S1. Reference-set processing

The ARROW reference table was reconstructed from machine-readable project outputs and contains 85 unique neutral solutes in water at 298 K. One molecule–condition row was retained for each connectivity. The reference includes experimental hydration free energy and, where available in the source, uncertainty, classical ARROW, PIMD4 and PIMD8 values. Molecular names were not used for identity. Canonical isomeric SMILES, full InChIKey and the first InChIKey block were generated with RDKit 2026.03.5. The released benchmark table is `data/benchmark/arrow_solvation_master.parquet`.

The reconstructed ARROW/PIMD8 MAE is 0.204836 kcal mol^{-1} and the classical ARROW MAE is 0.784648 kcal mol^{-1}. These values are direct errors against the same experimental column used for SolvAI evaluation; neither simulation value is an input to the final model.

S2. External endpoint labels

The endpoint-label catalog initially contained 5,075 benchmark-disjoint connectivities. A frozen source-quality rule retained all 1,147 rows from an earlier FreeSolv/CombiSolv-Exp merge and 133 additional SoluteML dGsolvDB3 rows supported by at least two source measurements, giving 1,280 external labels. Duplicate structures were resolved at the connectivity level before this rule was applied. The source composition is listed in Supplementary Table 3. The raw merged tables are not redistributed where source terms are not explicit; URLs, versions, hashes and reconstruction commands are given in `repro/DATA_PROVENANCE.md`.

For a random outer fold, these 1,280 rows were combined with the available ARROW outer-training molecules. External rows received sample weight 1 and ARROW outer-training rows weight 3. The experimental value of an outer-test molecule was never used. In the zero-ARROW transfer experiment, only the 1,280 external labels were fitted and all 85 ARROW molecules were evaluated.

The weight of 3 was part of the prospective confirmatory protocol rather than a post-result choice. A separately committed one-factor sensitivity changed only the available ARROW outer-training weight to 1. Under this setting, matched structure-only and full SolvAI MAEs are 0.30977 and 0.20642 kcal mol^{-1}, respectively; the paired change is -0.10335 kcal mol^{-1} (95% interval, -0.21883 to -0.01995).

S3. Response-source processing

CombiSolv-QM. The water subset was deduplicated by solute connectivity and expressed in kcal mol^{-1}. After the original exact-connectivity exclusion, 3,961 structures were available. Conservative canonical-tautomer comparison found two additional benchmark equivalents; the confirmatory teacher therefore used 3,959 rows.

SoluteML Abraham parameters. Five Abraham solute axes (E, S, A, B and L) were predicted from 8,098 benchmark-disjoint structures. These coordinates respectively describe excess molar refraction, dipolarity/polarizability, hydrogen-bond acidity, hydrogen-bond basicity and hexadecane–air partition response. No standardized benchmark equivalent was found in the audited source.

OpenFF explicit-water response. The source contains 520 benchmark-disjoint absolute solvation-free-energy records from an OpenFF 2.3.0/NAGL/TIP3P protocol with 14 alchemical states. Separate ExtraTrees models predicted calculated hydration free energy and its experimental residual. Their sum is the retained corrected OpenFF coordinate.

GBn2 implicit response. The source contains 550 benchmark-disjoint GBn2 hydration calculations. Separate ExtraTrees models predicted the calculated value and its experimental residual. The sum is the retained corrected implicit-solvent coordinate.

MolSolv SMD(water). The source archive contains 1,729,545 M06-2X/6-31G* SMD calculations. The frozen selection retained all eligible neutral structures with at most 10 heavy atoms and a deterministic one-in-eight sample for structures with 11–18 heavy atoms. After deduplication and exact-connectivity filtering, 350,391 rows remained. Canonical tautomer standardization identified 32 additional benchmark-equivalent records. The confirmatory model used 350,359 rows while keeping every retained molecule in its original source split.

ConfSolv H_2O. The archive contains 5,392,567 H_2O conformer records for 43,693 solutes. We converted kJ mol^{-1} to kcal mol^{-1} and aggregated conformers into gas, solution and hydration conformer corrections together with water-response distribution summaries. The broader quality filter produced 39,878 neutral connectivities. The selected teacher target set had 17,851 rows with complete features and admissible energy range; fragment-parent, uncharging and tautomer standardization removed 22 additional benchmark-equivalent records, leaving 17,829.

S4. The 15 response priors

Supplementary Table 1 defines every retained coordinate. At inference, the table's "surrogate" column is the calculation that runs; the expensive source method does not. The final vector contains one CombiSolv-QM response, five Abraham axes, one corrected OpenFF response, one corrected GBn2 response, one SMD(water) response and six ConfSolv summaries, for 15 coordinates in total.

The arithmetic is important. Richer CombiSolv, OpenFF, GBn2 and ConfSolv targets were available and were evaluated during model development. Only the listed coordinates belong to the released stack. In particular, no PIMD2, PIMD4, PIMD8, $\mathrm{d}H/\mathrm{d}\lambda$ or NQE coordinate is present.

S5. Surrogate models

The CombiSolv-QM and MolSolv targets were fitted with Chemprop 2.2.4 directed-message-passing models initialized from CHEMELEON. Both used a two-layer feed-forward head of width 256, dropout 0.1, initial/maximum/final learning rates of 3×10^{-5}, 10^{-4} and 10^{-5}, and PyTorch seed 20260826. CombiSolv-QM used RDKit 2D features, batch size 64, a 30-epoch cap and patience 7; MolSolv used the molecular graph, batch size 256, a 25-epoch cap and patience 6. The original frozen train/validation/test memberships were 80/10/10% for CombiSolv-QM and 90/5/5% for MolSolv. Standardized equivalents were removed from their existing split rather than resplitting the remaining source.

Abraham, OpenFF and GBn2 targets used ExtraTrees regressors on the same 2,265 structure features used by the endpoint. The selected leaf sizes and target-level OOF errors are recorded in the source metadata. ConfSolv used 217 RDKit descriptors and the 256 highest-variance Morgan bits. Each target was fitted with a LightGBM L1-regression model with 450 estimators, learning rate 0.04, 31 leaves, minimum child size 20, row subsampling 0.8, column subsampling 0.55 and L_2 regularization 0.1. Target-specific seeds were 20260826 plus the target index for validation and 1000 plus that index for the all-retained-source fit.

S6. Endpoint model

The molecular representation comprises a 2,048-bit radius-2 Morgan fingerprint and 217 RDKit 2D descriptors. The confirmatory endpoint appends the 15 response priors, giving 2,280 columns. A median imputer is fitted on each training partition and adds missingness indicators only where required. Each of three `ExtraTreesRegressor` models uses 360 trees, squared-error splitting, `max_features`=0.7, `min_samples_leaf`=2, `min_samples_split`=2, unlimited depth and no bootstrap; seeds are 11, 29 and 47. Predictions are averaged. The matched no-prior model removes only the 15 response columns.

After all evaluation was frozen, the release artifact was refitted on the 1,280 external labels and all 85 ARROW labels using the same weights and configuration. That refit supplies deployment predictions only. It is not used to calculate any reported held-out score.

S7. Validation partitions

The fixed primary result uses the released `fold_random` assignment. Five complete repeated five-fold partitions use shuffled K-fold seeds 314159, 271828, 161803, 141421 and 173205. All comparisons use the same fold, training rows and model seeds within a partition.

For functional-family separation, SMARTS rules were applied in the precedence order carboxylic acid, amide, ester, aldehyde, ketone, alcohol/phenol, ether, amine, thiol, sulfide/disulfide, aromatic heterocycle, carbocyclic aromatic, alkene/diene, saturated hydrocarbon and other. A test-fold family was removed from both public and ARROW endpoint training rows. Bemis–Murcko scaffold separation used canonical isomeric scaffold SMILES; acyclic molecules were keyed by primary family. Molecular clusters were constructed once on the combined endpoint pool with Taylor–Butina clustering at Morgan Tanimoto similarity 0.70. Nearest-neighbour analyses excluded training rows at similarity thresholds 0.50, 0.60, 0.70 and 0.80 to any outer-test molecule.

S8. Shuffled-prior control

All 15 response columns were permuted together to preserve their joint distribution. For each outer fold, public, ARROW outer-training and ARROW outer-test response rows were permuted separately. The experimental target was never used to define a permutation. Five base seeds (88001–88005) were used for the fixed partition and each repeated partition. This control changes molecule–response alignment while retaining feature count, response marginal distributions and inter-prior correlations.

S9. Statistical analysis

MAE is primary. RMSE, median absolute error, R^2 and the fraction of molecules improved are secondary. Paired candidate-minus-reference differences were resampled at molecule level 100,000 times with seed 20260828; percentile 2.5th and 97.5th quantiles define reported intervals. For the five repeated partitions, each molecule's absolute error was first averaged over partitions before paired resampling. The preregistration defined a positive comparison as a complete interval below zero and a material effect as an absolute MAE change of at least 0.010 kcal mol^{-1}. No correction was applied to convert the predeclared panel into one binary discovery claim; all comparisons are reported.

S10. Tier-A external validation

The original 221-molecule Tier-A cohort was frozen in a separate ASP-19/GB validation project before its model predictions. It was derived from type-M original-measurement records in Sander's Henry-law compilation v5.0.0. Intrinsic pure-water H_s^{cp} values normalized to 298.15 K were converted as $K_c = H_s^{cp}RT$ and $\Delta G_{\text{hyd}} = -RT \ln K_c$. This produces the same 1 M gas-to-1 M aqueous standard-state convention used for the endpoint. The source values are printed to two significant figures.

SolvAI qualification was performed without reading SolvAI predictions or errors. All 221 rows passed the connected, neutral, supported-element and target-compatibility rules. One candidate was excluded because the earlier ASP audit had documented a definite conflict between its registry name/CAS and supplied InChIKey structure. The remaining 220 molecules had no exact, fragment-parent, uncharged-parent or canonical-tautomer equivalent among the 1,280 external or 85 ARROW endpoint labels. Of these, 123 occurred in at least one supervised response-source table and 97 were absent from all six. The latter define the strict response-source-disjoint subset. The cohort hashes, all exclusions and all teacher-source matches were committed before endpoint evaluation.

Both external endpoints were then fitted deployment-style on the same 1,280 external plus 85 ARROW labels with weights 1 and 3. They used the same descriptors, imputation, three 360-tree ensembles and seeds; only the 15 response columns differed. No Tier-A label entered fitting or selection. On N=220, structure-only and SolvAI MAEs were 1.53165 and 1.15255 kcal mol^{-1} (paired change -0.37909; 95% interval, -0.51692 to -0.24817). On strict N=97, the corresponding values were 2.13830 and 1.53560 kcal mol^{-1} (paired change -0.60271; interval, -0.86277 to -0.35858). The absolute errors delimit applicability and are not presented as PIMD8-level performance outside ARROW-85.

S11. Lambda-response probes

Short PIMD2 response fingerprints were available at $\lambda = 0.1$, 0.5 and 0.9 for 72, 74 and 73 molecules, respectively; 72 had complete three-point curves. The targets were the total solute–solvent $\mathrm{d}H/\mathrm{d}\lambda$ and its electrostatic and van der Waals components. A zero-valued polarization field was not interpreted. Structure-only heads were evaluated by

OOF prediction, and their outputs were added to the same matched endpoint used for the corresponding campaign ablation. A separate direct-integration experiment used the predicted three-point curve with an affine calibration fitted inside each outer training fold. These targets were evaluated as candidate supervision and were not retained.

S12. Artifact verification

The public API accepts one or more SMILES strings. RDKit parses each input and emits canonical isomeric SMILES before feature generation. It does not standardize a query across tautomers, protonation states or salt forms. The runtime computes descriptors, evaluates the two D-MPNNs and four tree-artifact families, and passes the resulting 2,280-vector to three endpoint regressors. The artifact audit checks file hashes, feature count and names, software versions and two smoke predictions. It rejects features containing experimental target, PIMD, trajectory, probe, fold or scaffold information. The standardized-exclusion teacher refit is identified in the model card and manifest. The ensemble spread is returned for diagnosis but is not a calibrated per-query applicability or reliability score.

S13. Runtime benchmarking

Runtime was measured end to end from input SMILES to prediction on the released artifact. The host used an Intel Core i7-4930K CPU; the installed NVIDIA RTX 3090 was not used for inference. Cold single-molecule latency, five warm single-molecule runs, three batch-32 runs and peak resident memory were recorded. Chemprop predictor startup dominates single-molecule latency. These measurements describe the released implementation, not the irreducible model cost. The ARROW/PIMD8 comparison uses the published 15-window protocol and is kept separate from same-host inference timing.

Supplementary Notes

S1. Identity relationship to FreeSolv

Eighty of the 85 ARROW reference connectivities have a FreeSolv identity. This fact does not establish the provenance of every experimental value in the ARROW study. It instead determines the leakage rule: those connectivities were removed from all external supervised sources before endpoint or teacher fitting. The frozen SolvAI artifact is therefore not evaluated over unfiltered FreeSolv and no generic FreeSolv state-of-the-art claim is made.

S2. Interpretation of alternative supervision

Non-improving features are not interpreted as evidence that their underlying physical methods are inaccurate. A response can fail downstream because it is poorly predicted from structure, redundant with existing coordinates, mismatched to the endpoint or supported by too few labels. Compact ConfSolv summaries outperform larger graph and feed-forward latent representations in the matched exploratory screen. Sparse PIMD2 response produces physically recognizable trends but not the precision required for hydration-free-energy integration. These observations motivate the transfer condition stated in the main text: relevance, structural learnability and non-redundancy are all required.

Supplementary Tables

Supplementary Table 1 | SolvAI response priors

prior	name	physical meaning	source	units	surrogate	training rows	transformation	inference
1	combisolv_qm	COSMOtherm water response	CombiSolv-QM	kcal mol-1	CHEMELEON D-MPNN	3959	direct	structure surrogate; no simulation
2	abraham_e	excess molar refraction E	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
3	abraham_s	dipolarity/polarizability S	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
4	abraham_a	hydrogen-bond acidity A	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
5	abraham_b	hydrogen-bond basicity B	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
6	abraham_l	hexadecane-air partition L	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
7	openff_corrected	explicit-water alchemical response	OpenFF 2.3.0 ASFE	kcal mol-1	ExtraTrees	520	prediction + residual	structure surrogate; no simulation
8	gbn2_corrected	implicit-solvent hydration response	GBn2	kcal mol-1	ExtraTrees	550	prediction + residual	structure surrogate; no simulation
9	smd_water	SMD water response	MolSolv	kcal mol-1	CHEMELEON D-MPNN	350359	direct	structure surrogate; no simulation
10	conf_gas_corr	gas conformer correction	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
11	conf_solution_corr	solution conformer correction	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
12	conf_hydration_corr	hydration conformer correction	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
13	conf_gsolv_sd	conformer solvation-energy spread	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
14	conf_response_mean	mean conformer solvent response	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
15	conf_response_sd	conformer response spread	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation

Supplementary Table 2 | Source-specific confirmatory filtering

source	unit	before standardized exclusion	removed	retained	role
CombiSolv-QM	unique structures	3961	2	3959	COSMOtherm water
MolSolv	SMD calculations	350391	32	350359	SMD(water)
ConfSolv	model-usable connectivities	17851	22	17829	conformer response
Endpoint labels	experimental connectivities	1280	0	1280	hydration free energy

Supplementary Table 3 | External endpoint-label sources

source group	selected records	selection rule
FreeSolv only	69	benchmark-disjoint connectivity
CombiSolv-Exp only	588	water records; benchmark-disjoint connectivity
FreeSolv + CombiSolv-Exp	490	one duplicate-resolved connectivity
SoluteML dGsolvDB3	133	at least two source measurements

Supplementary Table 4 | Complete repeated-partition values

Repeat	Split seed	Method	MAE
0	314159	A_structure_only	0.31780
0	314159	F_full_solvai	0.20788
1	271828	A_structure_only	0.31059
1	271828	F_full_solvai	0.21010
2	161803	A_structure_only	0.31309
2	161803	F_full_solvai	0.21217
3	141421	A_structure_only	0.31602
3	141421	F_full_solvai	0.20618
4	173205	A_structure_only	0.30727
4	173205	F_full_solvai	0.20053

Supplementary Table 5 | Global chemical-separation results

regime	method	n	mae
global_family	A_structure_only	85	1.262
global_family	F_full_solvai	85	0.468
global_scaffold	A_structure_only	85	0.728
global_scaffold	F_full_solvai	85	0.376
global_butina_0_70	A_structure_only	85	0.349
global_butina_0_70	F_full_solvai	85	0.216
global_nn_0.50	A_structure_only	85	0.359
global_nn_0.50	F_full_solvai	85	0.219
global_nn_0.60	A_structure_only	85	0.344
global_nn_0.60	F_full_solvai	85	0.211
global_nn_0.70	A_structure_only	85	0.343
global_nn_0.70	F_full_solvai	85	0.211
global_nn_0.80	A_structure_only	85	0.339
global_nn_0.80	F_full_solvai	85	0.210

Supplementary Table 6 | Diagnostic family errors

family	n	mae_kcal_mol
Acids	3	0.230
Alcohols	13	0.196
Aldehydes	3	0.117
Alkanes	13	0.234
Alkenes	5	0.085
Amides	4	0.483
Amines	5	0.138
Aromatics	13	0.313
Esters	4	0.055
Ethers	4	0.241
Heterorings	5	0.122
Ketones	5	0.160
Sulfides	4	0.115
Thiols	4	0.095

Supplementary Table 7 | Release-artifact hashes

file	sha256	bytes
MODEL_CARD.md	a20bd031f5b85ba4f8fea701fa74f3d71f092ce79abb0c19b517711ff8ecfbde	1617
abraham_teacher.joblib	8c8a6365e7162dbe6ae445ce4004928ed86b2669d3e7a70d3d054f143ca97554	87055391
combisolv_qm.pt	2950be1232e2f868136e3388cf139e671d4196922ece0d3c7043d9dadece8b40	37452324
confsolv_teacher.joblib	a71487e53d5085e897dad02e62e95562933bc47262145fbf44286aae04b70b0d	5729365
head.joblib	fdc88cbae31696f254d85eb427b3c53a4811f29b3290d605b4f1ea66ebb95790	29424285
implicit_teacher.joblib	484d5e0c1efa759c8e7acbfcbbd87e218b93ae239e7f35303fe4cd7e310910f8	36316890
model_card.json	0f78177a701320a5aad77c45304ea73d5ca4799fb91e4f4fd01d0bb83f8e1244	1745
molsolv_smd.pt	35601030f030af72a9e6990bf8003df9d481858b27994b3fc70c81d605d130be	37226928
openff_teacher.joblib	55b7d31a293dd61fe57f5505e015775d7d86f53dca69980d398843706d58bb9b	48211661

Supplementary Table 8 | Runtime measurements

Measurement	Value
cold single molecule	15.312 s
warm single molecule median	15.294 s
warm single molecule p95	15.322 s
batch 32 median	15.824 s
batch 32 per molecule	0.494 s

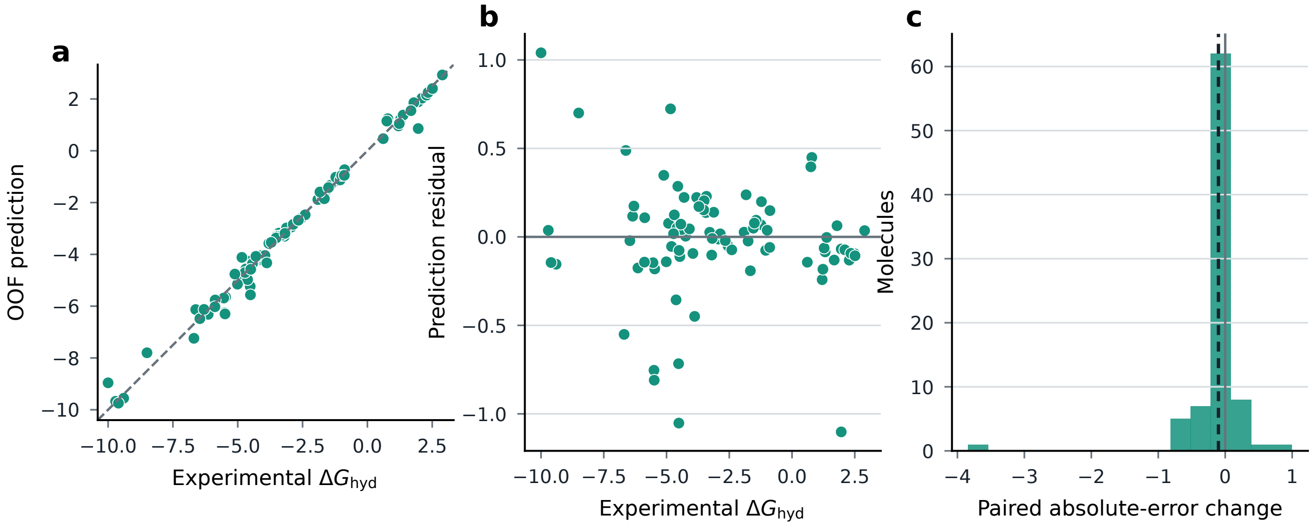
Supplementary Table 9 | Equal-weight ARROW sensitivity

Method	N	MAE	RMSE	Median absolute error
Matched structure-only	85	0.30977	0.59585	0.14972
Full SolvAI	85	0.20642	0.32627	0.11653

Supplementary Table 10 | Tier-A external validation

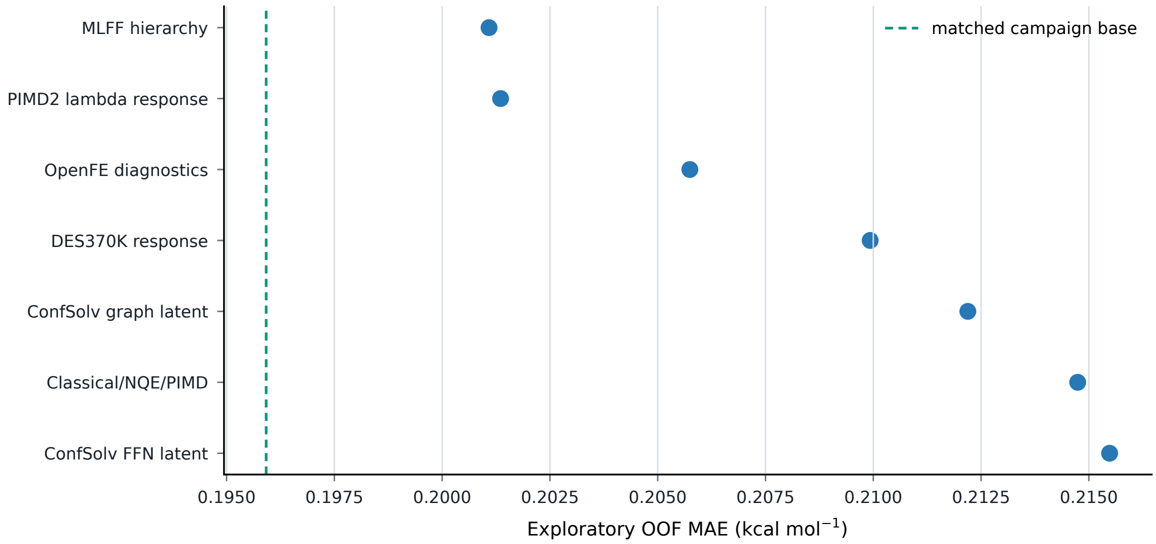
Cohort	Method	N	MAE	RMSE	Median absolute error
Endpoint-disjoint	Matched structure-only	220	1.53165	2.30974	0.96182
Endpoint-disjoint	Full SolvAI	220	1.15255	1.57723	0.84508
Strict response-source-disjoint	Matched structure-only	97	2.13830	3.08051	1.10237
Strict response-source-disjoint	Full SolvAI	97	1.53560	1.98792	1.24392

Supplementary Figures

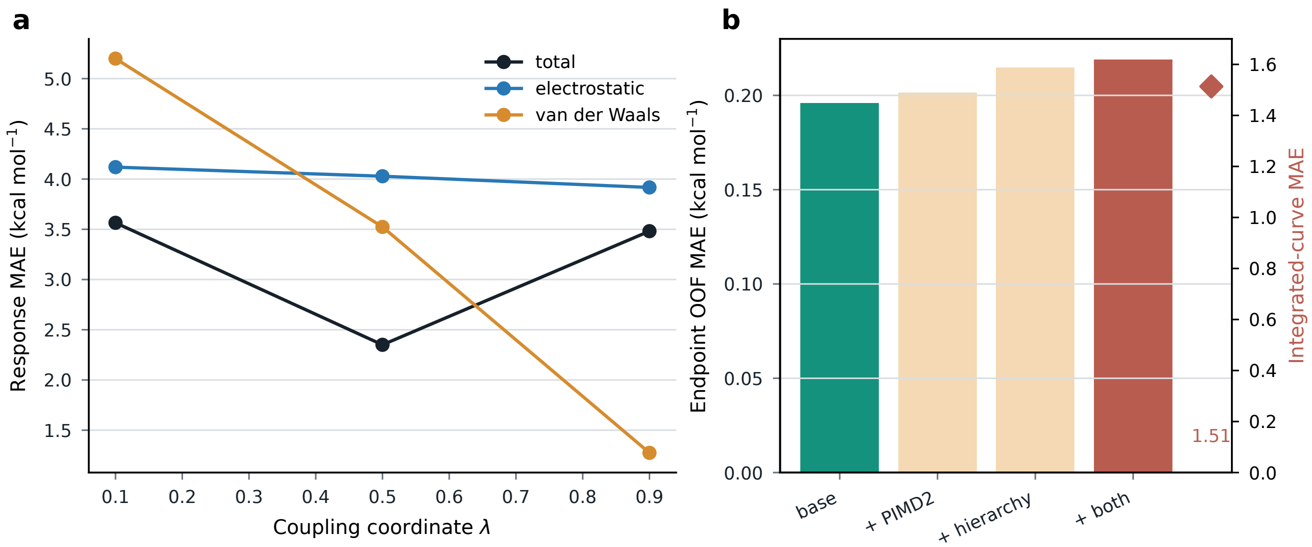


Supplementary Figure 1 | Molecule-level prediction and residuals. **a**, Confirmatory SolvAI OOF prediction against experiment. **b**, Signed residual against experimental hydration free energy. **c**, Distribution of the paired absolute-error change relative to the matched structure-only endpoint.

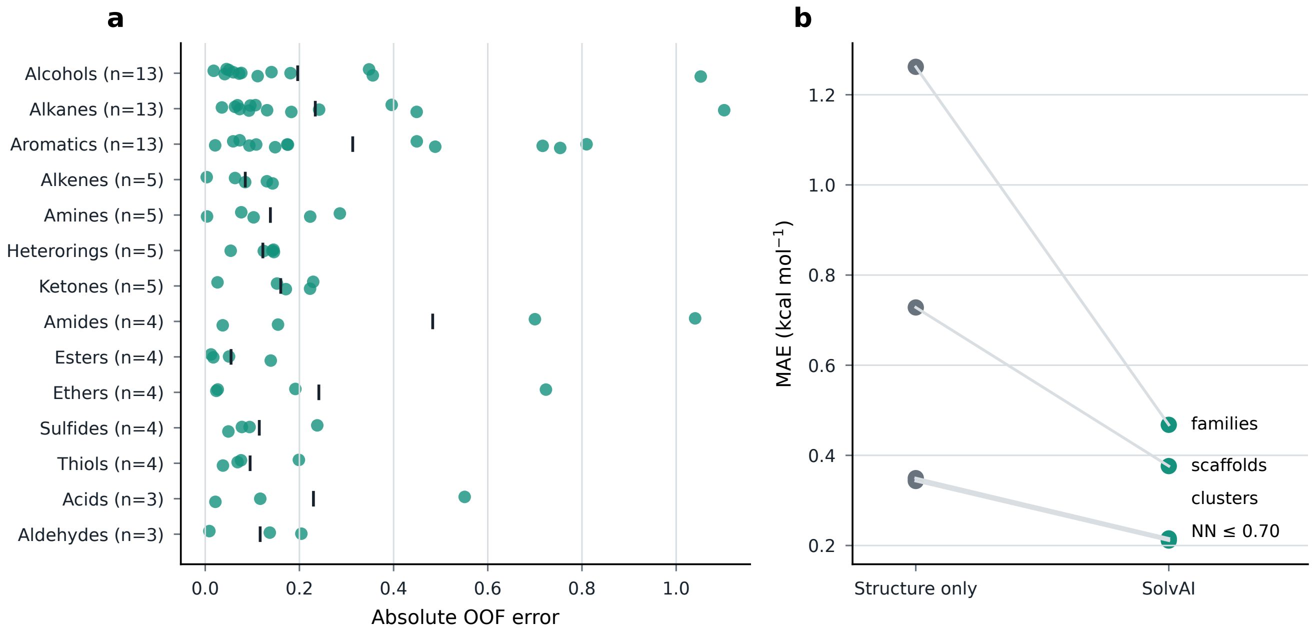
Supplementary Figure 2 | Source-specific provenance and standardized identity exclusion. Counts are shown in their native source units and normalized only within source. The audit removed standardized benchmark equivalents before the confirmatory teacher refits; no endpoint experimental label was removed.



Supplementary Figure 3 | Alternative physical-supervision routes. Exploratory fixed screens of additional diagnostic, latent and hierarchy features. The dashed line is the matched campaign response base for those screens. These experiments were not used to select the confirmatory controls.



Supplementary Figure 4 | Multi-λ response experiment. a, OOF response-head error at three coupling states. **b**, Matched endpoint consequences and the separately scaled error obtained by direct integration of the predicted curve.



Supplementary Figure 5 | Molecular and chemical-separation errors. **a**, Individual fixed-partition SolvAI errors grouped by diagnostic chemical family; ticks show family means and labels give sample sizes. **b**, Matched structure-only and SolvAI errors under four global separation regimes.